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\\MARTINOS DEVELOPER

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\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\VASO_POPE_apr_0.85s_0.95
_BOLD

TA: 3:06 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 16 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	90
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
Base Resolution	210
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	4
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	6/8
Slice Partial Fourier	Off
Asymmetric Echo	Weak

Routine

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
TR	32.4 ms
Vol. TR	2032 ms
TE 1	11.20 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
TR	32.4 ms
Vol. TR	2032 ms
Multi-echo Shots	1

Contrast - Common

TR	32.4 ms
Vol. TR	2032 ms
TE 1	11.20 ms
Multi-echo spacing	29.50 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	497.60 ms
TI 2	1502.00 ms
Flip Angle	30 deg
Fat-Water Contrast	Water Excitation
Contrasts	1
Reconstruction	Magnitude

Geometry - AutoAlign

Slab Group	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	R1.6 A11.8 F9.3
R	1.6 mm
A	11.8 mm
F	9.3 mm
Initial Orientation	T > C
T > C	-22.50
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.9 A5.3 F1.0 mm
! Orientation	T > C-17.9 > S0.4
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	163 mm
! F >> H	87 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136683 MHz
I Ref. Amplitude 1H	380.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso bb4d852
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Low SAR
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1488 Hz/Px
Echo Spacing	0.72 ms
Asymmetric Echo	Weak
Turbo Factor	31
Segmentation	1
EPI Factor	40

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	360 us
RF time x BW	5
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Expert ramp shaping	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI ramp factor	1.10
EPI ramp factor 2	1.20
Ramp sampling	1.00 1:max
Slab Scale	0 %
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off

Sequence - Special

GRAPPA Regularization	5000 /10^6
Spoiling scheme	Conventional
Read polarity	Regular RO
Pre-T11 delay	0.00 ms
Pre-T12 delay	0 ms
Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\VASO_POPE_apr_0.85s_0.95
_water_ext_underlay

TA: 2:16 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 16 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	50
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	200
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	4
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	6/8
Slice Partial Fourier	Off
Asymmetric Echo	Off

Routine

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	34.1 ms
Vol. TR	2637.4 ms
TE 1	12.20 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	34.1 ms
Vol. TR	2637.4 ms
Multi-echo Shots	1

Contrast - Common

TR	34.1 ms
Vol. TR	2637.4 ms
TE 1	12.20 ms
Multi-echo spacing	30.70 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	1023.10 ms
TI 2	2080.20 ms
Flip Angle	30 deg
Fat-Water Contrast	Water Excitation
Contrasts	1
Reconstruction	Magnitude

Geometry - AutoAlign

Slab Group	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	R1.6 A11.8 F9.3
R	1.6 mm
A	11.8 mm
F	9.3 mm
Initial Orientation	T > C
T > C	-22.50
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.9 A5.3 F1.0 mm
! Orientation	T > C-17.9 > S0.4
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	163 mm
! F >> H	87 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136683 MHz
! Ref. Amplitude 1H	380.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso bb4d852
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Low SAR
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1562 Hz/Px
Echo Spacing	0.79 ms
Asymmetric Echo	Off
Turbo Factor	31
Segmentation	1
EPI Factor	38

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	360 us
RF time x BW	5
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Expert ramp shaping	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI ramp factor	1.10
EPI ramp factor 2	1.20
Ramp sampling	1.00 1:max
Slab Scale	0 %
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off

Sequence - Special

GRAPPA Regularization	5000 /10^6
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Pre-T11 delay	500.00 ms
Pre-T12 delay	0 ms
Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\localizer

TA: 15 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	All Segments
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
TE	3.69 ms
Averages	2
Concatenations	3
AutoAlign	---

Contrast - Common

TR	8.6 ms
TE	3.69 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	20 deg

Contrast - Common

Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	256
Phase Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal

Geometry - Common

Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	3

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up

System - Adjustments

B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.136683 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8.6 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	250 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	20 deg
Measurements	1
Contrasts	1
TE	3.69 ms
TR	8.6 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	qfl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	320 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\VASO_POPE_apr_0.85s_0.95_noassym

TA: 13 sec Coil Selection: Manual Voxel Size: 1.0x1.0x1.0 mm³ Acc:: 16 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	4
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	200
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	4
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	6/8
Slice Partial Fourier	Off
Asymmetric Echo	Off

Routine

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	34.2 ms
Vol. TR	2143.6 ms
TE 1	12.30 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	34.2 ms
Vol. TR	2143.6 ms
Multi-echo Shots	1

Contrast - Common

TR	34.2 ms
Vol. TR	2143.6 ms
TE 1	12.30 ms
Multi-echo spacing	31.50 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	524.60 ms
TI 2	1584.80 ms
Flip Angle	45 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Geometry - AutoAlign

Slab Group	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	R1.6 A11.8 F9.3
R	1.6 mm
A	11.8 mm
F	9.3 mm
Initial Orientation	T > C
T > C	-22.50
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Transversal
! Rotation	0.00 deg
! A >> P	263 mm
! R >> L	350 mm
! F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136683 MHz
I Ref. Amplitude 1H	380.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso bb4d852
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Optimized
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1470 Hz/Px
Echo Spacing	0.81 ms
Asymmetric Echo	Off
Turbo Factor	31
Segmentation	1
EPI Factor	38

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1440 us
RF time x BW	25
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Expert ramp shaping	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI ramp factor	1.10
EPI ramp factor 2	1.20
Ramp sampling	1.00 1:max
Slab Scale	0 %
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off

Sequence - Special

GRAPPA Regularization	5000 /10^6
Spoiling scheme	Conventional
Read polarity	Regular RO
Pre-T11 delay	0.00 ms
Pre-T12 delay	0 ms
Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\scout_axial_princes

TA: 17 sec Coil Selection: Manual Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
Graphic segment	3rd Segment
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A16.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	160
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.600 mm
TR	3.6 ms
TE	1.56 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	3.6 ms
TE	1.56 ms
MTC	Off
Magn. Preparation	None
Flip Angle	16 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.600 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A16.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	160
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.600 mm
TR	3.6 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A16.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A16.0 H0.0
L	0.0 mm
A	16.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 A36.7 F31.6 mm
! Orientation	Transversal
! Rotation	0.00 deg
! A >> P	263 mm
! R >> L	350 mm
! F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.136683 MHz
! Ref. Amplitude 1H	250.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3.6 ms
Segments	1
Concatenations	1

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	260 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	16 deg
Measurements	1
Contrasts	1

Inline - MapIt

TE	1.56 ms
TR	3.6 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	540 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	AC

Sequence - Special

Readout polarity	Positive
Image processing	Standard
Apply echo spacing	Off
Echo spacing	0 us
Delta echo spacing	0 us
Dummy scans	0 ms
RF pulse duration	100 us
Gradient spoiling	Siemens
Gradient moment factor	1.00
Receiver gain mode	Siemens
Number of segments	1
Current segment	0
Lines before/after seg	0

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\scout_sag_princes

TA: 13 sec Coil Selection: Manual Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A16.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.600 mm
TR	3.5 ms
TE	1.51 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	3.5 ms
TE	1.51 ms
MTC	Off
Magn. Preparation	None
Flip Angle	16 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.600 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Allowed
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A16.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.600 mm
TR	3.5 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A16.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A16.0 H0.0
L	0.0 mm
A	16.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 A36.7 F31.6 mm
! Orientation	Transversal
! Rotation	0.00 deg
! A >> P	263 mm
! R >> L	350 mm
! F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.136683 MHz
! Ref. Amplitude 1H	250.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3.5 ms
Segments	1
Concatenations	1

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	260 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	16 deg
Measurements	1
Contrasts	1

Inline - MapIt

TE	1.51 ms
TR	3.5 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	540 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	AC

Sequence - Special

Readout polarity	Positive
Image processing	Standard
Apply echo spacing	Off
Echo spacing	0 us
Delta echo spacing	0 us
Dummy scans	0 ms
RF pulse duration	100 us
Gradient spoiling	Siemens
Gradient moment factor	1.00
Receiver gain mode	Siemens
Number of segments	1
Current segment	0
Lines before/after seg	0

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\scout_cor_princes

TA: 13 sec Coil Selection: Manual Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A21.6 F1.2 mm
Orientation	Coronal
Phase Encoding Dir.	R >> L
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.600 mm
TR	3.5 ms
TE	1.51 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	3.5 ms
TE	1.51 ms
MTC	Off
Magn. Preparation	None
Flip Angle	16 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.600 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Allowed
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A21.6 F1.2 mm
Orientation	Coronal
Phase Encoding Dir.	R >> L
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.600 mm
TR	3.5 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A21.6 F1.2 mm
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A21.6 F1.2
L	0.0 mm
A	21.6 mm
F	1.2 mm
Initial Orientation	Coronal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 A36.7 F31.6 mm
! Orientation	Transversal
! Rotation	0.00 deg
! A >> P	263 mm
! R >> L	350 mm
! F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.136683 MHz
! Ref. Amplitude 1H	250.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3.5 ms
Segments	1
Concatenations	1

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	260 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	16 deg
Measurements	1
Contrasts	1

Inline - MapIt

TE	1.51 ms
TR	3.5 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	540 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	AC

Sequence - Special

Readout polarity	Positive
Image processing	Standard
Apply echo spacing	Off
Echo spacing	0 us
Delta echo spacing	0 us
Dummy scans	0 ms
RF pulse duration	100 us
Gradient spoiling	Siemens
Gradient moment factor	1.00
Receiver gain mode	Siemens
Number of segments	1
Current segment	0
Lines before/after seg	0

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\wip_ShimToolbox_B0map_3
mm

TA: 44 sec Coil Selection: Auto Voxel Size: 3.0x3.0x3.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	15.0 ms
TE 1	2.20 ms
TE 2	5.40 ms
Concatenations	1
AutoAlign	---

Contrast - Common

TR	15.0 ms
TE 1	2.20 ms
TE 2	5.40 ms
MTC	Off
Magn. Preparation	None
Flip Angle	8 deg
Fat-Water Contrast	Water Excitation
Dark Blood	Off
Contrasts	2
SWI	Off
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
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Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
Base Resolution	72
Phase Resolution	100 %
Slice Resolution	100 %

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	15.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	216 mm
F >> H	216 mm
R >> L	156 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.136683 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off

System - Tx/Rx

Image Scaling	1.000
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Physio - Signal

1st Signal/Mode	None
TR	15.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Water Excitation
Magn. Preparation	None
Dark Blood	Off
FOV Read	216 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
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Inline - MapIt

Flip Angle	8 deg
Measurements	1
Contrasts	2
TE 1	2.20 ms
TE 2	5.40 ms
TR	15.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	WIP_GRE
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Readout Mode	Monopolar
Gradient Mode	Whisper
Flow Compensation 1	None
Flow Compensation 2	None
Bandwidth 1	500 Hz/Px
Bandwidth 2	500 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Mode	B0 Mapping
Masking	Brain Extraction
Phase Unwrapping	Spatial

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\VASO_POPE_apr_0.85s_0.95
_water_ext

TA: 12 sec Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 16 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	4
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	200
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	4
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	6/8
Slice Partial Fourier	Off
Asymmetric Echo	Off

Routine

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	34.1 ms
Vol. TR	2137.4 ms
TE 1	12.20 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	34.1 ms
Vol. TR	2137.4 ms
Multi-echo Shots	1

Contrast - Common

TR	34.1 ms
Vol. TR	2137.4 ms
TE 1	12.20 ms
Multi-echo spacing	30.70 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	523.10 ms
TI 2	1580.20 ms
Flip Angle	45 deg
Fat-Water Contrast	Water Excitation
Contrasts	1
Reconstruction	Magnitude

Geometry - AutoAlign

Slab Group	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	R1.6 A11.8 F9.3
R	1.6 mm
A	11.8 mm
F	9.3 mm
Initial Orientation	T > C
T > C	-22.50
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.9 A5.3 F1.0 mm
! Orientation	T > C-17.9 > S0.4
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	163 mm
! F >> H	87 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136683 MHz
! Ref. Amplitude 1H	380.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso bb4d852
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Low SAR
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1562 Hz/Px
Echo Spacing	0.79 ms
Asymmetric Echo	Off
Turbo Factor	31
Segmentation	1
EPI Factor	38

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	360 us
RF time x BW	5
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Expert ramp shaping	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI ramp factor	1.10
EPI ramp factor 2	1.20
Ramp sampling	1.00 1:max
Slab Scale	0 %
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off

Sequence - Special

GRAPPA Regularization	5000 /10^6
Spoiling scheme	Conventional
Read polarity	Regular RO
Pre-T11 delay	0.00 ms
Pre-T12 delay	0 ms
Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\VASO_POPE_apr_0.85s_0.95
_water_ext

TA: 12 sec Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 16 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	4
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	200
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	4
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	6/8
Slice Partial Fourier	Off
Asymmetric Echo	Off

Routine

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	34.1 ms
Vol. TR	2137.4 ms
TE 1	12.20 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	34.1 ms
Vol. TR	2137.4 ms
Multi-echo Shots	1

Contrast - Common

TR	34.1 ms
Vol. TR	2137.4 ms
TE 1	12.20 ms
Multi-echo spacing	30.70 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	523.10 ms
TI 2	1580.20 ms
Flip Angle	45 deg
Fat-Water Contrast	Water Excitation
Contrasts	1
Reconstruction	Magnitude

Geometry - AutoAlign

Slab Group	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	R1.6 A11.8 F9.3
R	1.6 mm
A	11.8 mm
F	9.3 mm
Initial Orientation	T > C
T > C	-22.50
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.9 A5.3 F1.0 mm
! Orientation	T > C-17.9 > S0.4
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	163 mm
! F >> H	87 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136683 MHz
! Ref. Amplitude 1H	380.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso bb4d852
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Low SAR
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1562 Hz/Px
Echo Spacing	0.79 ms
Asymmetric Echo	Off
Turbo Factor	31
Segmentation	1
EPI Factor	38

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	360 us
RF time x BW	5
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Expert ramp shaping	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI ramp factor	1.10
EPI ramp factor 2	1.20
Ramp sampling	1.00 1:max
Slab Scale	0 %
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off

Sequence - Special

GRAPPA Regularization	5000 /10^6
Spoiling scheme	Conventional
Read polarity	Regular RO
Pre-T11 delay	0.00 ms
Pre-T12 delay	0 ms
Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\VASO_POPE_apr_0.85s_0.95
_water_ext_VASO

TA: 3:35 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 16 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	100
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	200
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	4
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	6/8
Slice Partial Fourier	Off
Asymmetric Echo	Off

Routine

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	33.6 ms
Vol. TR	2106.4 ms
TE 1	12.10 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	33.6 ms
Vol. TR	2106.4 ms
Multi-echo Shots	1

Contrast - Common

TR	33.6 ms
Vol. TR	2106.4 ms
TE 1	12.10 ms
Multi-echo spacing	30.70 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	515.60 ms
TI 2	1557.20 ms
Flip Angle	45 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Geometry - AutoAlign

Slab Group	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	R1.6 A11.8 F9.3
R	1.6 mm
A	11.8 mm
F	9.3 mm
Initial Orientation	T > C
T > C	-22.50
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.9 A5.3 F1.0 mm
! Orientation	T > C-17.9 > S0.4
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	163 mm
! F >> H	87 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136683 MHz
! Ref. Amplitude 1H	380.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso bb4d852
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Low SAR
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1562 Hz/Px
Echo Spacing	0.79 ms
Asymmetric Echo	Off
Turbo Factor	31
Segmentation	1
EPI Factor	38

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	360 us
RF time x BW	5
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Expert ramp shaping	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI ramp factor	1.10
EPI ramp factor 2	1.20
Ramp sampling	1.00 1:max
Slab Scale	0 %
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off

Sequence - Special

GRAPPA Regularization	5000 /10^6
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Pre-T11 delay	0.00 ms
Pre-T12 delay	0 ms
Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\VASO_POPE_apr_0.85s_0.95
_water_ext

TA: 13 sec Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 16 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	4
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	200
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	4
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	6/8
Slice Partial Fourier	Off
Asymmetric Echo	Off

Routine

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	34.1 ms
Vol. TR	2137.4 ms
TE 1	12.20 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	34.1 ms
Vol. TR	2137.4 ms
Multi-echo Shots	1

Contrast - Common

TR	34.1 ms
Vol. TR	2137.4 ms
TE 1	12.20 ms
Multi-echo spacing	30.70 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	523.10 ms
TI 2	1580.20 ms
Flip Angle	45 deg
Fat-Water Contrast	Water Excitation
Contrasts	1
Reconstruction	Magnitude

Geometry - AutoAlign

Slab Group	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	R1.6 A11.8 F9.3
R	1.6 mm
A	11.8 mm
F	9.3 mm
Initial Orientation	T > C
T > C	-22.50
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.9 A5.3 F1.0 mm
! Orientation	T > C-17.9 > S0.4
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	163 mm
! F >> H	87 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136683 MHz
! Ref. Amplitude 1H	380.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso bb4d852
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Low SAR
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1562 Hz/Px
Echo Spacing	0.79 ms
Asymmetric Echo	Off
Turbo Factor	31
Segmentation	1
EPI Factor	38

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	360 us
RF time x BW	5
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Expert ramp shaping	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI ramp factor	1.10
EPI ramp factor 2	1.20
Ramp sampling	1.00 1:max
Slab Scale	0 %
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off

Sequence - Special

GRAPPA Regularization	5000 /10^6
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Pre-T11 delay	0.00 ms
Pre-T12 delay	0 ms
Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\VASO_POPE_apr_0.85s_0.95
_water_ext_underlay

TA: 1:50 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 16 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	50
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	200
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	4
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	6/8
Slice Partial Fourier	Off
Asymmetric Echo	Off

Routine

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	34.1 ms
Vol. TR	2137.4 ms
TE 1	12.20 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	34.1 ms
Vol. TR	2137.4 ms
Multi-echo Shots	1

Contrast - Common

TR	34.1 ms
Vol. TR	2137.4 ms
TE 1	12.20 ms
Multi-echo spacing	30.70 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	523.10 ms
TI 2	1580.20 ms
Flip Angle	30 deg
Fat-Water Contrast	Water Excitation
Contrasts	1
Reconstruction	Magnitude

Geometry - AutoAlign

Slab Group	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	R1.6 A11.8 F9.3
R	1.6 mm
A	11.8 mm
F	9.3 mm
Initial Orientation	T > C
T > C	-22.50
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.9 A5.3 F1.0 mm
! Orientation	T > C-17.9 > S0.4
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	163 mm
! F >> H	87 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136683 MHz
! Ref. Amplitude 1H	380.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso bb4d852
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Low SAR
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1562 Hz/Px
Echo Spacing	0.79 ms
Asymmetric Echo	Off
Turbo Factor	31
Segmentation	1
EPI Factor	38

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	360 us
RF time x BW	5
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Expert ramp shaping	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI ramp factor	1.10
EPI ramp factor 2	1.20
Ramp sampling	1.00 1:max
Slab Scale	0 %
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off

Sequence - Special

GRAPPA Regularization	5000 /10^6
Spoiling scheme	Conventional
Read polarity	Regular RO
Pre-T11 delay	0.00 ms
Pre-T12 delay	0 ms
Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\VASO_POPE_apr_0.85s_0.95
_BOLD

TA: 1:45 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 16 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	50
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
Base Resolution	210
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	4
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	6/8
Slice Partial Fourier	Off
Asymmetric Echo	Weak

Routine

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
TR	32.4 ms
Vol. TR	2032 ms
TE 1	11.20 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
TR	32.4 ms
Vol. TR	2032 ms
Multi-echo Shots	1

Contrast - Common

TR	32.4 ms
Vol. TR	2032 ms
TE 1	11.20 ms
Multi-echo spacing	29.50 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	497.60 ms
TI 2	1502.00 ms
Flip Angle	30 deg
Fat-Water Contrast	Water Excitation
Contrasts	1
Reconstruction	Magnitude

Geometry - AutoAlign

Slab Group	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	R1.6 A11.8 F9.3
R	1.6 mm
A	11.8 mm
F	9.3 mm
Initial Orientation	T > C
T > C	-22.50
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.9 A5.3 F1.0 mm
! Orientation	T > C-17.9 > S0.4
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	163 mm
! F >> H	87 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136683 MHz
! Ref. Amplitude 1H	380.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso bb4d852
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Low SAR
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1488 Hz/Px
Echo Spacing	0.72 ms
Asymmetric Echo	Weak
Turbo Factor	31
Segmentation	1
EPI Factor	40

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	360 us
RF time x BW	5
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Expert ramp shaping	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI ramp factor	1.10
EPI ramp factor 2	1.20
Ramp sampling	1.00 1:max
Slab Scale	0 %
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off

Sequence - Special

GRAPPA Regularization	5000 /10^6
Spoiling scheme	Conventional
Read polarity	Regular RO
Pre-T11 delay	0.00 ms
Pre-T12 delay	0 ms
Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\VASO_POPE_apr_0.85s_0.95
_BOLD

TA: 3:06 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 16 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	90
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
Base Resolution	210
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	4
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	6/8
Slice Partial Fourier	Off
Asymmetric Echo	Weak

Routine

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
TR	32.4 ms
Vol. TR	2032 ms
TE 1	11.20 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
TR	32.4 ms
Vol. TR	2032 ms
Multi-echo Shots	1

Contrast - Common

TR	32.4 ms
Vol. TR	2032 ms
TE 1	11.20 ms
Multi-echo spacing	29.50 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	497.60 ms
TI 2	1502.00 ms
Flip Angle	30 deg
Fat-Water Contrast	Water Excitation
Contrasts	1
Reconstruction	Magnitude

Geometry - AutoAlign

Slab Group	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	R1.6 A11.8 F9.3
R	1.6 mm
A	11.8 mm
F	9.3 mm
Initial Orientation	T > C
T > C	-22.50
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.9 A5.3 F1.0 mm
! Orientation	T > C-17.9 > S0.4
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	163 mm
! F >> H	87 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136683 MHz
! Ref. Amplitude 1H	380.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso bb4d852
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Low SAR
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1488 Hz/Px
Echo Spacing	0.72 ms
Asymmetric Echo	Weak
Turbo Factor	31
Segmentation	1
EPI Factor	40

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	360 us
RF time x BW	5
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Expert ramp shaping	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI ramp factor	1.10
EPI ramp factor 2	1.20
Ramp sampling	1.00 1:max
Slab Scale	0 %
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off

Sequence - Special

GRAPPA Regularization	5000 /10^6
Spoiling scheme	Conventional
Read polarity	Regular RO
Pre-T11 delay	0.00 ms
Pre-T12 delay	0 ms
Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\VASO_POPE_apr_0.85s_0.95
_BOLD

TA: 3:06 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 16 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	90
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
Base Resolution	210
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	4
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	6/8
Slice Partial Fourier	Off
Asymmetric Echo	Weak

Routine

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
TR	32.4 ms
Vol. TR	2032 ms
TE 1	11.20 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
TR	32.4 ms
Vol. TR	2032 ms
Multi-echo Shots	1

Contrast - Common

TR	32.4 ms
Vol. TR	2032 ms
TE 1	11.20 ms
Multi-echo spacing	29.50 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	497.60 ms
TI 2	1502.00 ms
Flip Angle	30 deg
Fat-Water Contrast	Water Excitation
Contrasts	1
Reconstruction	Magnitude

Geometry - AutoAlign

Slab Group	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	R1.6 A11.8 F9.3
R	1.6 mm
A	11.8 mm
F	9.3 mm
Initial Orientation	T > C
T > C	-22.50
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.9 A5.3 F1.0 mm
! Orientation	T > C-17.9 > S0.4
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	163 mm
! F >> H	87 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136683 MHz
! Ref. Amplitude 1H	380.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso bb4d852
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Low SAR
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1488 Hz/Px
Echo Spacing	0.72 ms
Asymmetric Echo	Weak
Turbo Factor	31
Segmentation	1
EPI Factor	40

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	360 us
RF time x BW	5
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Expert ramp shaping	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI ramp factor	1.10
EPI ramp factor 2	1.20
Ramp sampling	1.00 1:max
Slab Scale	0 %
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off

Sequence - Special

GRAPPA Regularization	5000 /10^6
Spoiling scheme	Conventional
Read polarity	Regular RO
Pre-T11 delay	0.00 ms
Pre-T12 delay	0 ms
Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\VASO_POPE_apr_0.85s_0.95
_BOLD

TA: 3:06 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 16 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	90
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
Base Resolution	210
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	4
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	6/8
Slice Partial Fourier	Off
Asymmetric Echo	Weak

Routine

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
TR	32.4 ms
Vol. TR	2032 ms
TE 1	11.20 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
TR	32.4 ms
Vol. TR	2032 ms
Multi-echo Shots	1

Contrast - Common

TR	32.4 ms
Vol. TR	2032 ms
TE 1	11.20 ms
Multi-echo spacing	29.50 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	497.60 ms
TI 2	1502.00 ms
Flip Angle	30 deg
Fat-Water Contrast	Water Excitation
Contrasts	1
Reconstruction	Magnitude

Geometry - AutoAlign

Slab Group	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	R1.6 A11.8 F9.3
R	1.6 mm
A	11.8 mm
F	9.3 mm
Initial Orientation	T > C
T > C	-22.50
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.9 A5.3 F1.0 mm
! Orientation	T > C-17.9 > S0.4
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	163 mm
! F >> H	87 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136683 MHz
! Ref. Amplitude 1H	380.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso bb4d852
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Low SAR
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1488 Hz/Px
Echo Spacing	0.72 ms
Asymmetric Echo	Weak
Turbo Factor	31
Segmentation	1
EPI Factor	40

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	360 us
RF time x BW	5
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Expert ramp shaping	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI ramp factor	1.10
EPI ramp factor 2	1.20
Ramp sampling	1.00 1:max
Slab Scale	0 %
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off

Sequence - Special

GRAPPA Regularization	5000 /10^6
Spoiling scheme	Conventional
Read polarity	Regular RO
Pre-T11 delay	0.00 ms
Pre-T12 delay	0 ms
Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\VASO_POPE_apr_0.85s_0.95
_BOLD

TA: 3:06 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 16 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	90
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
Base Resolution	210
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	4
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	6/8
Slice Partial Fourier	Off
Asymmetric Echo	Weak

Routine

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
TR	32.4 ms
Vol. TR	2032 ms
TE 1	11.20 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
TR	32.4 ms
Vol. TR	2032 ms
Multi-echo Shots	1

Contrast - Common

TR	32.4 ms
Vol. TR	2032 ms
TE 1	11.20 ms
Multi-echo spacing	29.50 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	497.60 ms
TI 2	1502.00 ms
Flip Angle	30 deg
Fat-Water Contrast	Water Excitation
Contrasts	1
Reconstruction	Magnitude

Geometry - AutoAlign

Slab Group	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	R1.6 A11.8 F9.3
R	1.6 mm
A	11.8 mm
F	9.3 mm
Initial Orientation	T > C
T > C	-22.50
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.9 A5.3 F1.0 mm
! Orientation	T > C-17.9 > S0.4
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	163 mm
! F >> H	87 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136683 MHz
! Ref. Amplitude 1H	380.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso bb4d852
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Low SAR
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1488 Hz/Px
Echo Spacing	0.72 ms
Asymmetric Echo	Weak
Turbo Factor	31
Segmentation	1
EPI Factor	40

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	360 us
RF time x BW	5
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Expert ramp shaping	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI ramp factor	1.10
EPI ramp factor 2	1.20
Ramp sampling	1.00 1:max
Slab Scale	0 %
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off

Sequence - Special

GRAPPA Regularization	5000 /10^6
Spoiling scheme	Conventional
Read polarity	Regular RO
Pre-T11 delay	0.00 ms
Pre-T12 delay	0 ms
Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\HelloFromTheOtherSide\20240429_SP04\VASO_POPE_apr_0.85s_0.95
_BOLD_OR

TA: 3:06 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 16 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	90
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
Base Resolution	210
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	4
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	6/8
Slice Partial Fourier	Off
Asymmetric Echo	Weak

Routine

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
TR	32.4 ms
Vol. TR	2032 ms
TE 1	11.20 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0.0 %
Slice Oversampling	3.3 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	0.95 mm
TR	32.4 ms
Vol. TR	2032 ms
Multi-echo Shots	1

Contrast - Common

TR	32.4 ms
Vol. TR	2032 ms
TE 1	11.20 ms
Multi-echo spacing	29.50 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	497.60 ms
TI 2	1502.00 ms
Flip Angle	30 deg
Fat-Water Contrast	Water Excitation
Contrasts	1
Reconstruction	Magnitude

Geometry - AutoAlign

Slab Group	1
Position	R1.6 A11.8 F9.3 mm
Orientation	T > C-22.5
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	R1.6 A11.8 F9.3
R	1.6 mm
A	11.8 mm
F	9.3 mm
Initial Orientation	T > C
T > C	-22.50
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.9 A5.3 F1.0 mm
! Orientation	T > C-17.9 > S0.4
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	163 mm
! F >> H	87 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136683 MHz
! Ref. Amplitude 1H	380.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso bb4d852
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Low SAR
Gradient Mode	Fast
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Bandwidth	1488 Hz/Px
Echo Spacing	0.72 ms
Asymmetric Echo	Weak
Turbo Factor	31
Segmentation	1
EPI Factor	40

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	360 us
RF time x BW	5
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Expert ramp shaping	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI ramp factor	1.10
EPI ramp factor 2	1.20
Ramp sampling	1.00 1:max
Slab Scale	0 %
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off

Sequence - Special

GRAPPA Regularization	5000 /10^6
Spoiling scheme	Conventional
Read polarity	Regular RO
Pre-T11 delay	0.00 ms
Pre-T12 delay	0 ms
Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off
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